Project: Worked examples
Title: Thick-walled cylinder (Lamé)
Status: DRAFT · Author: TempestOS (2026-10-23) · Checker: — (—) · Issued: —

1. Scope and method

Radial, hoop and axial stresses at the bore and the outer surface of a thick-walled cylinder under internal and external pressure, the von Mises and Tresca equivalents, utilisation against an allowable, and the thin-wall estimate for comparison.

WHAT IT CALCULATES	RELATION
Lamé constants from the bore and outside pressures	$A = (a^2 p_i - b^2 p_o)/(b^2 - a^2)$; $B = a^2 b^2 (p_i - p_o)/(b^2 - a^2)$
Radial and hoop stress at radius r	$\sigma_r = A - B/r^2$; $\sigma_\theta = A + B/r^2$
Axial stress, closed or open ends	$\sigma_z = A$ (closed); $\sigma_z = 0$ (open)
Equivalent stresses from the three principal stresses	von Mises; Tresca = the largest principal difference

Basis: Thick-walled cylinders[1](Ch. 13, Table 13.5); Stresses in pressurised cylinders[2](§3-14).

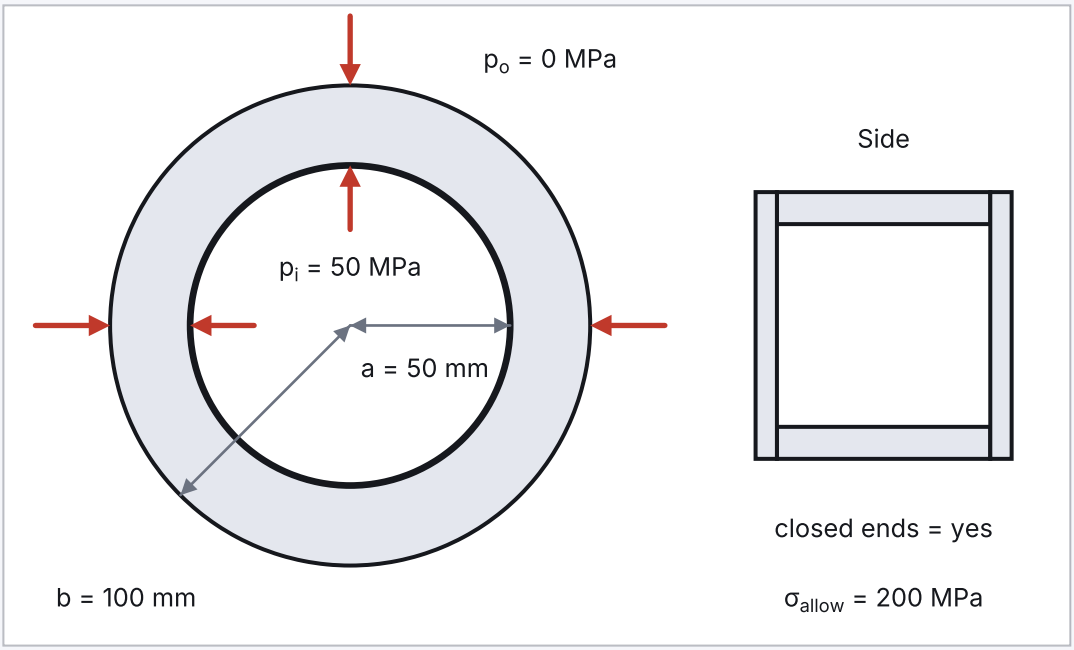


Figure 1 — A thick-walled cylinder in section, bore radius a, outer radius b, pressure inside and out; closed ends carry the pressures axially. Not to scale; the labels are the inputs as entered.

2. Inputs

SYMBOL	DESCRIPTION	VALUE	UNIT	SOURCE	VALID RANGE
a	Inner radius	50	mm	drawing: DRG-CYL-01 rev A	0.1 … 5000 mm
b	Outer radius	100	mm	drawing: DRG-CYL-01 rev A	0.1 … 5000 mm
p_i	Internal pressure	50	MPa	spec: SPEC-PV-2 §3 design pressure	0 … 1000 MPa

SYMBOL	DESCRIPTION	VALUE	UNIT	SOURCE	VALID RANGE
p_o	External pressure	0	MPa	assumption: Atmospheric outside.	0 … 1000 MPa
closed	Closed ends (axial stress carried by the wall)	yes		drawing: DRG-CYL-01 end caps rev A	any
E	Young's modulus (the elastic collapse pressure under external pressure only)	207	GPa	default	1 … 500 GPa
v	Poisson's ratio (the elastic collapse pressure under external pressure only)	0.3		default	0.1 … 0.49
σ_{allow}	Allowable stress	200	MPa	spec: SPEC-PV-2 §5	1 … 3000 MPa

3. Assumptions

A1. Long cylinder, elastic, isotropic; stresses vary with radius only. (Lamé solution; end effects neglected.)
A2. Closed ends carry the pressure end load as a uniform axial stress. (σ_z = A.)
A3. p_o (External pressure) is taken as 0 MPa. (Atmospheric outside.)

4. Working

[9] A = (a^2 * p_i - b^2 * p_o) / (b^2 - a^2) [1](Table 13.5)
= ((50 mm)^2 × 50 MPa − (100 mm)^2 × 0 MPa) / ((100 mm)^2 − (50 mm)^2)
= 16.667 MPa
Lamé constant A

[10] B = a^2 * b^2 * (p_i - p_o) / (b^2 - a^2) [1](Table 13.5)
= (50 mm)^2 × (100 mm)^2 × (50 MPa − 0 MPa) / ((100 mm)^2 − (50 mm)^2)
= 166670 MPa·mm²
Lamé constant B

[11] $\sigma_{r,a}$ = A - B / a^2
= 16.667 MPa − 166670 MPa·mm² / (50 mm)^2
= -50 MPa
Radial stress at the bore

[12] $\sigma_{t,a}$ = A + B / a^2
= 16.667 MPa + 166670 MPa·mm² / (50 mm)^2
= 83.333 MPa
Hoop stress at the bore

[13] $\sigma_{r,b}$ = A - B / b^2
= 16.667 MPa − 166670 MPa·mm² / (100 mm)^2
= 1.8626e-15 MPa
Radial stress at the outer surface

[14] $\sigma_{t,b}$ = A + B / b^2
= 16.667 MPa + 166670 MPa·mm² / (100 mm)^2
= 33.333 MPa
Hoop stress at the outer surface

[15] σ_z = A
= 16.667 MPa
= 16.667 MPa
Axial stress (closed ends)

[16] $\sigma_{vm,a}$ = sqrt(((($\sigma_{t,a}$ - $\sigma_{r,a}$)^2 + ($\sigma_{r,a}$ - σ_z)^2 + (σ_z - $\sigma_{t,a}$)^2) / 2)
= sqrt(((83.333 MPa − (-50 MPa))^2 + ((-50 MPa) − 16.667 MPa)^2 + (16.667 MPa − 83.333 MPa)^2

= 115.47 MPa

von Mises stress at the bore

[17] $\sigma_{vm,b} = \text{sqrt}(((\sigma_{t,b} - \sigma_{r,b})^2 + (\sigma_{r,b} - \sigma_z)^2 + (\sigma_z - \sigma_{t,b})^2) / 2)$

= sqrt(((33.333 MPa - 1.8626e-15 MPa)^2 + (1.8626e-15 MPa - 16.667 MPa)^2 + (16.667 MPa - 33.333 MPa)^2) / 2)

= 28.868 MPa

von Mises stress at the outer surface

[18] $\sigma_{tr,a} = \max(\sigma_{t,a}, \sigma_{r,a}, \sigma_z) - \min(\sigma_{t,a}, \sigma_{r,a}, \sigma_z)$

= max(83.333 MPa, -50 MPa, 16.667 MPa) - min(83.333 MPa, -50 MPa, 16.667 MPa)

= 133.33 MPa

Tresca stress at the bore

[19] $\sigma_{tr,b} = \max(\sigma_{t,b}, \sigma_{r,b}, \sigma_z) - \min(\sigma_{t,b}, \sigma_{r,b}, \sigma_z)$

= max(33.333 MPa, 1.8626e-15 MPa, 16.667 MPa) - min(33.333 MPa, 1.8626e-15 MPa, 16.667 MPa)

= 33.333 MPa

Tresca stress at the outer surface

[20] $\sigma_{vm,max} = \max(\sigma_{vm,a}, \sigma_{vm,b})$

= max(115.47 MPa, 28.868 MPa)

= 115.47 MPa

Maximum von Mises stress

[21] $u = \sigma_{vm,max} / \sigma_{allow}$

= 115.47 MPa / 200 MPa

= 0.57735

Utilisation

[22] $\sigma_{thin} = (p_i - p_o) * (a + b) / 2 / (b - a)$

= (50 MPa - 0 MPa) × (50 mm + 100 mm) / 2 / (100 mm - 50 mm)

= 75 MPa

Thin-wall hoop estimate at the mean radius

[23] $k_{wall} = b / a$

= 100 mm / 50 mm

= 2

Wall ratio b/a

5. Results

SYMBOL	VALUE	DESCRIPTION
$\sigma_{r,a}$	-50 MPa	Radial stress, bore
$\sigma_{t,a}$	83.33 MPa	Hoop stress, bore
$\sigma_{r,b}$	0 MPa	Radial stress, outer
$\sigma_{t,b}$	33.33 MPa	Hoop stress, outer
σ_z	16.67 MPa	Axial stress
$\sigma_{vm,a}$	115.5 MPa	von Mises, bore
$\sigma_{vm,b}$	28.87 MPa	von Mises, outer
$\sigma_{tr,a}$	133.3 MPa	Tresca, bore
$\sigma_{tr,b}$	33.33 MPa	Tresca, outer
$\sigma_{vm,max}$	115.5 MPa	Maximum von Mises
σ_{thin}	75 MPa	Thin-wall hoop estimate
k_{wall}	2	Wall ratio

SYMBOL	VALUE	DESCRIPTION
u	0.58	Utilisation (governing)

6. Checks (governing first)

RESULT	CHECK	DEMAND	CAPACITY	UTILISATION	REF
PASS	von Mises against allowable	u = 0.58	u ≤ 1	0.58	

7. Applicability

None.

Defaults used (inputs left empty):
E: not entered, so the default of 207 GPa is used.
v: not entered, so the default of 0.3 is used.

8. Verification

Template mech.thick-cylinder@2.1.1 Content hash aed5864571238a05 Engine calc-engine/1.0.0
Known-answer cases passed 3/3 Test run 2026-10-23 12:00 UTC TempestOS 0.26.0

9. References

1. Young WC, Budynas RG, Sadegh AM. Roark's Formulas for Stress and Strain. 8th ed. McGraw-Hill; 2012.
2. Budynas RG, Nisbett JK. Shigley's Mechanical Engineering Design. 11th ed. McGraw-Hill Education; 2020.